

MMM-NewLegoSets

THE COMPLETE 2.1 USER GUIDE

New LEGO sets. Your layout. Your colors. Your motion.

10 layouts | 11 themes | 23 animations | 11 indicator styles



SEPARATE CLOCKS

Poll LEGO.com every few hours while slides move every few seconds.

RECENT BY DESIGN

A 31-day window prioritizes exact official announcements and releases.

MIRROR READY

Responsive container layouts and reduced-motion support are built in.

CACHE RESCUE

A failed request keeps the last good product list visible and labeled.

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This manual accompanies the exhaustive README in the module folder. It focuses on setup, recent-product rules, visual choices, common recipes, and quick reference.

1. Install and start

Place the folder at **MagicMirror/modules/MMM-NewLegoSets**. There are no npm runtime dependencies. Add a module entry to MagicMirror's config and restart MagicMirror.

```
{
  module: "MMM-NewLegoSets",
  position: "top_right",
  config: {
    productCount: 1,
    layout: "hero",
    theme: "lego"
  }
}
```

What the first run does

1. FETCH

The Node helper requests LEGO.com's new-products listing.

2. NORMALIZE

Product image, price, pieces, number, and availability are normalized.

3. CACHE

A successful pool is stored as last-known-good data.

4. CYCLE

The browser displays 1-10 products and advances independently.

The independent clocks

Setting	Default	Meaning
data.pollInterval	6 hours	Download a fresh list from LEGO.com.
cycle.interval	12 seconds	Change the visible group in transition mode.
cycle.scrollSpeed	60 px/s	Move continuously in scroll mode without downloading.

Use milliseconds for polling and timed transitions. Scroll speed is measured in CSS pixels per second.

2. Choose a layout

The arrangements cover familiar product-display patterns. Auto is the safest starting point; it selects hero, list, or grid based on product count.

AUTO Hero for 1, list for 2-3, grid for 4-10.	HERO One large image/details presentation.
LIST Stacked image-and-details rows.	GRID Uniform multi-column product cards.
COMPACT Dense thumbnail rows for small regions.	SPLIT Wide image-led cards.
CAROUSEL Main product plus preview rail.	FILMSTRIP One horizontal sequence.
MASONRY Flowing columns for varied metadata.	TABLE Aligned, dense, highly scannable rows.

Sizing controls

Setting	Use
productCount	Visible products, 1 through 10.
layoutSettings.columns	Maximum grid and masonry columns, 1 through 10.
moduleWidth / moduleMaxWidth	Region width and maximum pixel width.
cardMinWidth	Preferred card width; caps columns without forcing overflow.
gap / padding / cardPadding	Density and breathing room.
imagePosition / alignItems	Card composition and vertical alignment.

3. Choose a theme

Every layout consumes the same theme variables, so switching a theme changes cards, text, accents, borders, background, and decorations together.

Theme	Palette	Decoration
mirror	Black / white	None
lego	Red / yellow / blue	Corner bricks
tuxedo	Black / silver / white	Bow tie
pride	Rainbow	Rainbow bars
progress	Progress Pride colors	Chevron
trans	Blue / pink / white	Flag bars
bisexual	Pink / purple / blue	Flag bars
lesbian	Orange / white / pink	Flag bars
nonbinary	Yellow / white / purple / black	Flag bars
pansexual	Pink / yellow / cyan	Flag bars
custom	Your values	Selectable



Set `showThemeDecorations: false` to keep a preset's palette and remove its corner artwork. Use theme: "custom" to control background, surface, text, muted, accent, accent2, border, shadow, gradient, and decoration.

4. Motion and LEGO build

Twenty-three choices cover restrained fades through a literal full-module brick wall. The module honors the operating system's reduced-motion preference by default.

SUBTLE	DIRECTIONAL
none, fade, crossfade, blur	slideLeft, slideRight, slideUp, slideDown
DEPTH	PLAYFUL
zoomIn, zoomOut, flipX, flipY	rotate, roll, bounce, swing, elastic
REVEALS	BRICKS
wipe, shutter	legoBuild, legoBreakBuild, brickWallRebuild, random

```
animation: {
  name: "brickWallRebuild",
  duration: 1000,
  easing: "cubic-bezier(0.22, 1, 0.36, 1)",
  stagger: 18,
  particleCount: 36,
  brickSize: 10,
  wallColumns: 6,
  wallRows: 5,
  respectReducedMotion: true
}
```

The full brick wall

brickWallRebuild clips the complete rendered module - header, image, text, indicators, footer, and decorations - into a wall. The outgoing wall falls below the module; the incoming wall descends from above and resolves into one solid module.

Leave animation.brickColors empty to use the active theme palette. Supply 1-16 hex colors to override it. The trans theme defaults to #5BCEFA, #F5A9B8, and #FFFFFF.

Performance recipe

For Raspberry Pi-class mirrors, start with fade, 600-800 ms, and 12-24 particles. For brickWallRebuild, try a 4 x 3 or 5 x 4 wall before the 6 x 5 default. High slice counts, blur, and large shadows increase GPU work.

5. Data, cycling, and caching

Data key	Default	Explanation
locale / countryCode	en-us / US	Storefront, formatting, and Brickset region.
pageCount	2	Listing pages attempted, 1-8.
poolSize	10	Up to 50; page mode reserves two full pages.
coming soon / preorders	true / true	Include announced and preorder products.
recentOnly / recentDays	true / 31	Enable and size the recent window.
unknownDatePolicy	firstSeen	Fallback, include, or strict exclude.
newsroomAnnouncements	true	Match official article dates by set number.
pollInterval	6 hours	Successful refresh clock.
cacheEnabled / max age	true / 7 days	Persist and reuse the last good pool.
sortBy / direction	recent / desc	Exact announcements first and newest first.

Cycle key	Default	Explanation
enabled	true	Automatic slide movement.
mode	transition	Timed effects or continuous scroll.
interval	12 seconds	Delay between transition-mode changes.
scrollSpeed	60 px/s	Continuous speed, 10-500.
scrollDirection	left	Left, right, up, or down.
step	page	Replace the complete visible group.
loop	true	Wrap at the end.
shuffle	false	Shuffle each fresh pool.
indicatorStyle	dots	Dots, rings, shapes, bars, numbers, or none.

Auto-scroll support

Scroll mode is available for hero, carousel, and filmstrip. Other layouts use timed transitions. `animation.name` and `cycle.interval` are ignored while continuous scrolling is active.

Page-sized slideshow

```
productCount: 2,  
data: { poolSize: 10, pollInterval: 6 * 60 * 60 * 1000 },  
cycle: { enabled: true, interval: 12 * 1000, step: "page" }
```

With one product, page advances by one. With two products, both are replaced. Numeric steps 1-10 remain available for intentional overlapping windows.

6. Configure every field

Available fields: image, name, price, pieceCount, setNumber, releaseDate, announcedDate, availability, ageRange, and pricePerPiece.

Image controls

Key	Meaning
show	Show or hide the product image.
width / height	Preferred box size, default 180 x 180 px.
minWidth / maxWidth	Responsive limits.
aspectRatio	CSS aspect ratio, default 1 / 1.
fit / position	Contain, cover, fill, scale-down and alignment.
opacity	0 is transparent; 1 is opaque.
background / borderRadius	Image stage appearance.
filter	Any valid CSS filter.
showPlaceholder	LEGO-colored fallback if an image is absent.

Text controls

Keys	Meaning
show, order	Visibility and ordering.
label, prefix, suffix	Value wording.
missingText	Fallback; empty hides unavailable data.
fontFamily, fontSize, fontWeight, fontStyle	Field-specific typography.
letterSpacing, lineHeight, lineClamp	Text density and truncation.
textTransform, textAlign	Case and alignment.
color, opacity	Field color/transparency.
effect	none, shadow, outline, glow, neon, gradient, or letterpress.

7. Dates without guesswork

The module deliberately preserves unknowns. This prevents plausible-looking but false product history.

LEGO LISTING

Explicit available/release/launch dates may be used.

NEWSROOM

Official article dates are matched to exact set numbers.

BRICKSET

Optional regional dateFirstAvailable or launchDate enrichment.

FIRST SEEN

A transparent fallback for new-listing products with no exact date.

```
data: {
  recentOnly: true,
  recentDays: 31,
  unknownDatePolicy: "firstSeen",
  newsroomAnnouncements: true,
  newsroomPageLimit: 30,
  sortBy: "recent",
  sortDirection: "desc"
}
```

Exact announcements sort first, then exact past releases, then first-seen fallbacks. Shipping and backorder estimates are ignored. First-seen time is never displayed as an announcement or release date. Choose unknownDatePolicy: "exclude" when only exact dates are acceptable.

8. Ready-made recipes

Ten-product Progress Pride grid

```
layout: "grid",
productCount: 10,
theme: "progress",
data: { poolSize: 20 },
cycle: { interval: 30000, step: "page", indicatorStyle: "bars" },
layoutSettings: { columns: 2, moduleMaxWidth: 900 }
```

Tuxedo information table

```
layout: "table",
productCount: 8,
theme: "tuxedo",
animation: { name: "wipe", duration: 700 },
typography: { baseFontSize: 14 },
fields: {
  releaseDate: { show: false },
  announcedDate: { show: false },
  availability: { show: false }
}
```

Transparent images, no motion

```
layout: "list",
productCount: 4,
theme: "mirror",
cycle: { enabled: false },
animation: { name: "none" },
fields: { image: { opacity: 0.55, filter: "saturate(.8)" } }
```

Continuous star filmstrip

```
layout: "filmstrip",
productCount: 5,
cycle: {
  mode: "scroll",
  scrollSpeed: 70,
  scrollDirection: "left",
  indicatorStyle: "stars"
}
```

Full wall fall and rebuild

```
layout: "hero",
productCount: 1,
theme: "trans",
cycle: { mode: "transition", interval: 15000 },
animation: {
  name: "brickWallRebuild",
  duration: 1000,
  stagger: 18,
  brickColors: [],
  wallColumns: 6,
  wallRows: 5
}
```

9. Validation and troubleshooting

Run the offline validator after every config edit. Run live validation after updates or when LEGO.com output looks wrong.

```
cd ~/MagicMirror/modules/MMM-NewLegoSets
npm run validate
npm run validate:live
```

Symptom	Check
Only one set	Raise productCount to show more; or keep 1 and ensure poolSize > 1 for cycling.
Sets never change	cycle.enabled, mode, interval/speed, and poolSize larger than productCount.
Images blank	Access to www.lego.com/cdn and DNS filtering.
Dates blank	No exact source matched; choose fallback/strict policy, Brickset, or overrides.
Motion missing	The operating system may request reduced motion.
Too many symbols	Keep step page, switch to dots/bars, or hide indicators.
Scroll will not start	Use hero, carousel, or filmstrip and keep cycling enabled.
Narrow overflow	Use auto/compact/table, fewer columns, or larger cardMinWidth.
Custom font missing	Use a reachable WOFF2 path, preferably local.
Brickset warning	Check key/connectivity; products still render without enrichment.
Listing parser error	Run validate:live and inspect MagicMirror's terminal.

Validated release matrix

5,584 pairwise normalized states and 1,249 rendered editor states; all 10 layouts, 11 themes, 23 animations, 11 indicators, product counts 1-10, fields, text effects, device presets, and boundaries. All 69 two-product hero/carousel/filmstrip transition lifecycles and all 12 continuous-scroll layout/direction cases passed. A live 31-day feed check returned 47 unique canonical set numbers and exact Newsroom announcements newest-first on August 16, 2026.

10. Quick reference

Top-level defaults

Key	Default
title / subtitle	New LEGO Sets / Recently released
layout / productCount	auto / 1
theme / decorations	lego / true
dateFormat	year numeric, month short, day numeric
debug	false

Layout defaults

Key	Default
columns / gap	2 / 12
cardMinWidth	170
moduleWidth / maxWidth	auto / 760
padding / cardPadding	12 / 10
borderRadius / cardBorderRadius	8 / 6
imagePosition / alignItems	left / center
header / footer / counter / updated	all true
source	false

Typography defaults

Key	Default
fontFamily	Roboto, Arial, sans-serif
customFontName / Url	empty
baseFontSize / align	16 / left
textEffect / antialias	none / true

Keep building

The module is designed to be changed. Start with product count, layout, and theme. Then tune cycle timing and animation. Finally, adjust fields and typography once the information density feels right on the physical mirror.

START SIMPLE

auto + one product + your preferred theme

CHOOSE MOTION

cycle.mode selects timed transitions or continuous scroll

PRESERVE UNKNOWNNS

Blank dates are better than invented history

VALIDATE

Run both validators before relying on an upgrade

Documentation and community references

MagicMirror core module, rendering, and animation documentation; Brickset API v3 documentation; official LEGO Newsroom; MMM-Carousel; and MagicMirror community discussion of CSS animation performance. Full links are in README.md.

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